

Guide

CUSTOMER

SAP Information Collaboration Hub for Life Sciences

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Regulatory Reporting for the EU-Hub

SAP Information Collaboration Hub for Life Sciences



Document History

Version	Date	Status	Change
1.0	15-Feb-18	Released to POC customers	Initial release
1.1	24-Feb-18	Released to POC customers	Updates: • Graphic • Section numbers added to Functions
1.2	4-May-18	Released to customers	POC specifics removed.
1.3	6-Dec-2018	Final	Updates: • Support for new interface version • Alerts

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1 About this Document

1.1 Purpose and Scope

This document describes the service provided by the SAP Information Collaboration Hub for Life Sciences for Regulatory Reporting to the European Medicines Verification Organization (EMVO).

The document also describes the procedure customers must follow to use the service.

1.2 Target Audience

This document is for customers and partners of the SAP Information Collaboration Hub for Life Sciences who use the EMVO Regulatory Reporting function, including:

- Implementation engineers
- IT security officers
- Quality Managers

Related Information

- [SAP Information Collaboration Hub for Life Sciences Help Portal](#)
- [EMVO Knowledge Base](#)

1.3 Acronyms

Acronym	Description
ATTP	SAP Advanced Track and Trace for Pharmaceuticals
EMVO	European Medicines Verification Organization
EMVS	European Medicines Verification System
Info Hub	SAP Information Collaboration Hub for Life Sciences, formerly Pharma Network
MAH	Marketing Authorization Holder
NMVS	National Medicines Verification System
OBP	On-boarding Partner

Acronym	Description
SSO	Single Sign On

2 Introduction

2.1 Background

According to the Falsified Medicines Directive (FMD), established by the European Union in 2013, Marketing Authorization Holders must report item level product information for any prescription drug intended to be sold within the EU market starting in February 2019. To facilitate this reporting, stakeholders set up the European Medicines Verification Organization (EMVO) to establish a system that supports the reporting of such data as well as verification at the point of dispense. This European Medicines Verification System (EMVS) consists of the European Hub managed by the EMVO and the various National Medicines Verification Systems (NMVS) managed by the relevant national organizations established for this purpose. Marketing Authorization Holders report data to the European Hub (EU-Hub).

The following sections describe the functions supported by the European Hub.

2.1.1 Product Service

The Product Service is used to receive requests and alerts related to product master and product pack data.

Process Product Service Connection Test

- Check if you are connected correctly to the product service: Validate authentication and encryption parameters.
- This can be used as an Onboarding Partner (OBP) operational tool for connectivity.

Process Product Service Subscription

- Registers an 'endpoint' on the requesting system to receive product-related notifications from the European Hub.
- A Marketing Authorization Holder must register with this service to receive notifications about product-related alerts.

Process Product Service Subscription Roll

- Activates the changes done by product service subscription request. Each user will have two endpoints and certificates registered with the EU Hub, one set as primary and one as secondary. The EU Hub will use the primary. Product service subscription roll request makes the endpoint and certificate primary.

Process Product Master Data

- This instructs the European Hub to distribute the supplied product master data to the relevant national system or systems.

Process Product Pack Data

- This instructs the European Hub to distribute the supplied product pack data to the relevant national system or systems. The operation is used for both sending new product pack data and for sending modified product packs data.

Process Batch Recall

- This instructs the European Hub to distribute a product batch recall request to a set of requested markets.

Process Product Pack State Update

- This instructs the European Hub to distribute a change of product pack state to one or more product packs. Clients operating on behalf of a manufacturer may set packs in the available state to 'decommissioned' and they may set packs in the decommissioned state to 'available' if they previously set the pack to 'decommissioned'.

Process Product Pack Verification

- This instructs the European Hub to initiate verification of a set of one or more product packs. This will return the actual state of each pack.

Process Product Withdrawal

This instructs the European Hub to distribute a product batch recall request to a set of requested markets.

2.1.2 Report Service

The reporting service is responsible for processing all requests related to system and business process reports.

Process Report Request

- The Process Report Request message instructs the European Hub to generate a report of the required type.

Get Report

- The Get Report message is a synchronous call used to retrieve a previously generated set of report data. The client supplies a report ID and a 'mime' type, and, if the requested report exists, the report is returned in the requested format.

2.1.3 Support Service

The ProcessSecuritySessionToken is called to register a new SecuritySessionToken. A security session token is a random string chosen by the caller. It must be included as a WS-Security token in every call made to any service on the European Hub manufacturer interface, in addition to the X.509 security token.

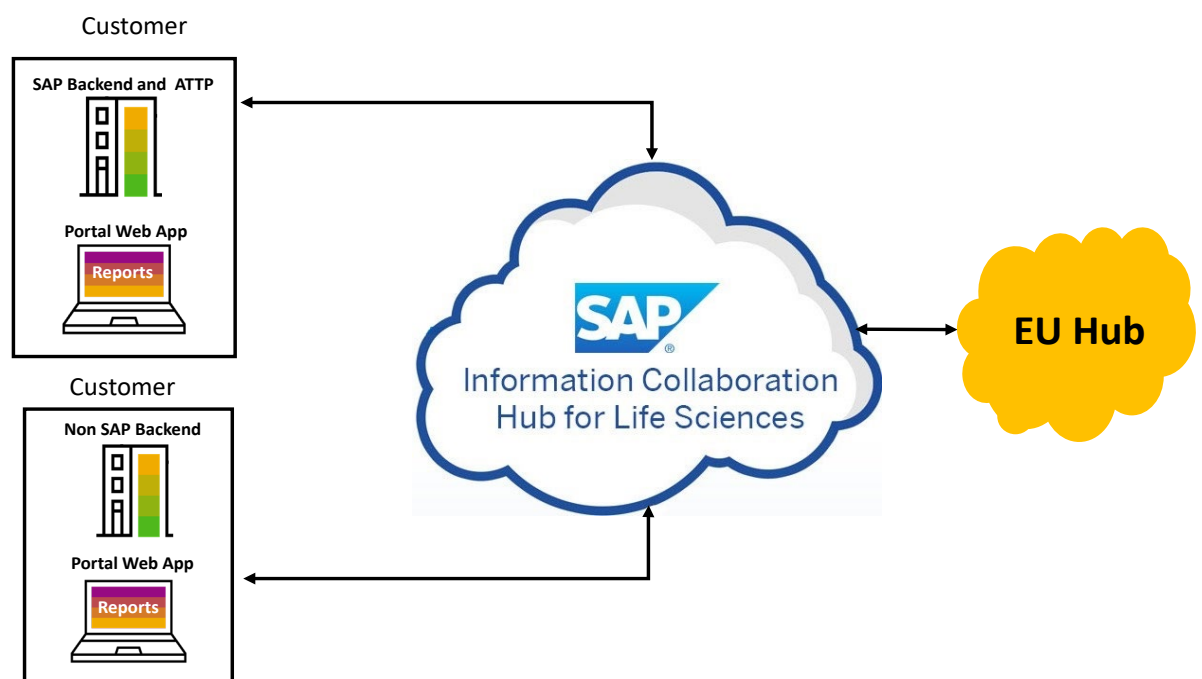
A SecuritySessionToken lifespan is chosen at time of creation. The lifespan must be at least 24 hours and may be as long as 120 hours. ProcessSecuritySessionToken is the only operation that will accept an expired security token, but this token must still match the latest token.

3 SAP EU-Hub Service Overview

The EU-Hub Regulatory Reporting service of the SAP Information Collaboration Hub for Life Sciences is available to customers who wish to connect:

- An SAP backend system such as SAP Advanced Track and Trace for Pharmaceuticals
- A non-SAP backend system
- Using the Web App to request reports

The following figure summarizes connection possibilities.



The SAP Information Collaboration Hub for Life Sciences routes messages to the EU-Hub received from various backend systems including the following features:

- Encryption and signing of messages
- Token handling
- Error handling
- Routing responses from EU-Hub back to the message sender

Note

Customers must send messages that comply with EMVO message format.

3.1 Key Benefits

The key benefits are as follows:

1. Facilitates the transmission of messages between Pharma MAHs and the EU Hub
2. Minimizes effort for setup, testing and activation of EU-Hub profiles
3. Accelerates connection to the EU-Hub
4. Standardizes communication to the EU-Hub
5. Dynamic connection to interface versions:
 - o 20161
 - o 20181

3.2 SAP Message Orchestration

The SAP Information Collaboration Hub for Life Sciences handles messages as follows:

1. Data reporting to EU Hub from backend for requests as well as routing to backend of related asynchronous status messages from EMVO system.
2. Alerts: messages originating in the EMVS are sent to appropriate cloud database and highlighted in the Reporting Web App
3. Report requests that originate from the Reporting Web App, and corresponding report responses are highlighted in the App.

3.3 Reporting Web App Features

The Reporting Web App allows users to request and receive reports from the EMVO. Features include:

- Report Request
Users can request reports from the EU-Hub based on predefined scenarios described and released by EMVO in PDF and CSV format.
- Report Download
Users can view the status of a requested report. The report is retrieved automatically, and a user can download the report.
- Report Admin
SAP manages configuration of the Request Report feature.
- Alerting
The App shows EMVO alerts including the related Alert ID and enables users to analyze those.

4 Applicable GxP Requirements

The SAP Information Collaboration Hub for Life Sciences is not a system of record, and therefore, can be regarded as a system with low risk towards GxP. Nevertheless, the data transmitted may be GxP relevant so basic GxP principles of developing and operating the system are applied.

GxP Factor	Status
Audit trail	n/a, data is not changed, logging is available for 30 days
Electronic Signatures	n/a
Security Requirements	For Routing: certificates expiring For Reporting App: - Password change required, length - User authentication - Logging and locking
Development	SAP Quality Management System applies
Operations	SAP Cloud Platform is SOC1, SOC2 and ISO 27001 certified.

4.1 Applicable Regulations and Requirements

Document	Date
False Medicines Directive (FMD)	09.Feb 2016
EMVS0714 - EMVS SDK for OBPs	31.Oct 2017
WSDL from EMVSJavaSDK_MAH • ManufacturerService.wsdl • ProductServiceMahV20161.wsdl • ReportingServiceMahV20161.wsdl	05.Feb 2018
WSDL from EMVSJavaSDK_MAH	24.Aug 2018

Document	Date
• CallbackService.wsdl • ProductServiceMahV20181.wsdl • ReportingServiceMahV20181.wsdl	

5 Functions

5.1 Main Functions

5.1.1 Product Master Data

The service accepts formatted messages for the product master data (initial upload and update) process for EMVO ProductMasterDataRequest.

The service accepts all corresponding responses from the EMVS-Hub (ProductMasterDataStatus) and sends information back to the requestor.

Reference EMVS714 Section 8.1, for details of the message structure, refer to the WSDL.

Messages can be sent according to interface version 20161 or 20181.

5.1.2 Product Pack

The service can accept formatted messages for the product pack data process for EMVO ProductPackDataRequest, "create" and "update".

The service can accept all corresponding responses from the EMVS-Hub (ProductPackDataStatus) and send information back to the requestor.

Reference EMVS714 Section 8.2, for details of message structure, refer to WSDL.

The message can be sent according to interface version 20161 or 20181.

5.1.3 Product Pack State Update

The service can accept formatted messages for the product pack state update process for EMVO ProductPackStateUpdateRequest.

The service can accept all corresponding responses from the EMVS-Hub (ProductPackStateUpdateStatus) and send information back to the requestor.

Reference EMVS714 Section 8.6, for details of message structure, refer to WSDL.

The message can be sent according to interface version 20161 or 20181.

5.1.4 Product Pack Verification

The service can accept formatted messages for the product pack verification process for EMVO ProductPackVerificationRequest.

The service can accept all corresponding responses from the EMVS-Hub (ProductPackStateVerificationStatus) and send information back to the requestor.

Reference EMVS714 Section 8.5, for details of the message structure, refer to the WSDL.

Messages can be sent according to interface version 20161 or 20181.

5.1.5 Batch Recall

The service can accept readily formatted messages for the process Batch Recall for EMVO ProductBatchRecallRequest.

The service can accept all corresponding responses from the EMVS-Hub (ProductBatchRecallStatus) and send information back to the requestor.

Reference EMVS714 Section 8.3, for details of message structure, refer to WSDL.

The message can be sent according to interface version 20161 or 20181.

5.1.6 Product Withdrawal

The service can accept formatted messages for Process Withdrawal for EMVO ProductWithdrawalRequest.

The service can accept all corresponding responses from the EMVS-Hub (ProductWithdrawalStatus) and send information back to the requestor.

Reference EMVS714 Section 8.4, for details of message structure, refer to WSDL.

Messages can be sent according to interface version 20161 or 20181.

5.1.7 Alerts and Reports

The service includes an application to support Alerts and Reports according to interface version 20181.

Alerts:

- ProductServiceSubscriptionRequest
- AlertRequest

Reference EMVS714 Section 8.8, 8.9, 8.10, for details of the message structure, refer to the WSDL.

Reports:

- ReportRequest
- ReportReady
- ReportRetrivalRequest

- ReportData

Reference EMVS714 Section 8.11, 8.12, for details of the message structure, refer to the WSDL.

5.1.8 Alert Requests

The SAP Information Collaboration Hub for Life Sciences EU Hub service receives AlertRequests from the EU Hub and displays the alerts in the Alert Web App.

Reference EMVS714 Section 8.9, for details of the message structure, refer to the WSDL.

Alerts are received in interface format 20181.

Web App users can:

- Filter alerts
- Set an alert to analyzed
- Delete an alert

5.1.9 Backwards Compatibility

EMVO supports backwards compatibility of interfaces. Users can submit requests to the EU Hub in both interface formats. The SAP Information Collaboration Hub for Life Sciences only supports the new interfaces for Alert, Report, and Token handling between ICH and EMVO.

You do not need to inform SAP when you switch backend system from interface 20161 to 20181 as SAP supports both interfaces.

SAP Information Collaboration Hub for Life Sciences recognizes the message format and routes the message according to the correct interface version, and corresponding URL.

5.1.10 Token Management and Encryption

The service manages the EMVS token and encryption process as specified by the EMVO.

Detailed information is described in EMVS714 Section 7.4 and 8.12. For details of the message structure, refer to the WSDL.

Token handling and token update are performed according to interface 20181. Token update is every 24 hours.

5.1.11 Communication Patterns

The service communicates with the EU-Hub synchronously and asynchronously. SOAP calls are processed as described in EMVS0714 Section 4.3 & 5.

The SAP Information Collaboration Hub for Life Sciences sends requests to the EU-Hub synchronously, and an acknowledgement is returned to the customer endpoint.

5.1.12 Access Management

The service can manage access to the OBP-Account for associated MAHs and CMOs including consideration of process and country.

5.1.13 Message Routing

The service supports routing of every EMVO message to the correct recipient.

5.1.14 Message Validation

The service ensures incoming messages are in the EU-Hub format using XML validation for both interface types.

5.1.15 Connection Test and Subscription Roll

The service provides an application to support the following which is performed during the onboarding process by SAP:

- Process Product Service Connection Test
Reference EMVS714 Section 8.7, for details of message structure, refer to the WSDL.
- Process Product Service Subscription Roll
Reference EMVS714 Section 8.8, for details of message structure, refer to the WSDL.

5.2 Performance and Timing Requirements

The service processes one million Serial numbers for Pack upload.

5.3 App Access

Apps include user management capabilities, including a username and password entry login process per GxP requirements. SAP sets up users.

5.3.1 Single Sign On

The service supports Single Sign On (SSO).

5.4 Alarms and Alarm Notifications

5.4.1 Exception Alarms

The service raises alarms in for exception events.

5.4.2 Alarm Notifications

The SAP Support team receives notifications in the event of exception events.

5.5 Technical Communication

5.5.1 AS2 Support

Using AS2, the service can:

- Collect files.
- Provide response files

5.5.2 HTTPS and ATTP

The service can communicate using HTTPS. SAP Advanced Track and Trace for Pharmaceuticals (ATTP) uses HTTPS.

6 Important Disclaimers and Legal Information

6.1 Coding Samples

Any software coding and/or code lines/strings ("Code") included in this documentation are only examples and are not intended to be used in a productive system environment. The Code is only intended to better explain and visualize the syntax and phrasing rules of certain coding. SAP does not warrant the correctness and completeness of the Code given herein, and SAP shall not be liable for errors or damages caused by the usage of the Code, unless damages were caused by SAP intentionally or by SAP's gross negligence.

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